MA653P - Computational Financial Modelling Lab
Assignment 3

Note: Use any library available in python.

1. **Gradient-Based Optimization with Multiple Restarts**
   (a) Implement a gradient-based method with multiple random restarts with 30 independent runs for each function.
   (b) Test on:
      i. Rastrigin
      $$\min_{\mathbf{x}\in[-5.12,5.12]^n} \quad f(\mathbf{x}) = 10n + \sum_{i=1}^{n} \left(x_i^2 - 10\cos(2\pi x_i)\right)$$
      Use $n = 2$.
      ii. Rosenbrock
      $$\min_{\mathbf{x}\in[-3,3]^n} \quad f(\mathbf{x}) = \sum_{i=1}^{n-1} \left[100(x_{i+1} - x_i^2)^2 + (1 - x_i)^2\right]$$
      Use $n = 2$.
   (c) Report:

Function	Best	Worst	Average	Std. Dev.
Rastrigin				
Rosenbrock				
Himmelblau				

2. **Binary Genetic Algorithm**
   (a) Implement Binary GA for the same functions.
   (b) Perform 30 runs and report:

Function	Best	Worst	Average	Std. Dev.
Rastrigin				
Rosenbrock				
Himmelblau				

3. **Parameter Sensitivity (Binary GA)**
   (a) Mutation probabilities: $0.01,\ 0.02,\ 0.05$
   (b) Crossover probabilities: $0.7,\ 0.8,\ 0.9$
   (c) Perform experiments (30 runs each) and plot error vs parameters.
   (d) Fill average error table:

	Crossover Probability		
Mutation	0.7	0.8	0.9
0.01			
0.02			
0.05			

4. **Real-Coded Genetic Algorithm** Consider the problem
   Optimization Problem
   (a) Solve:
   $$f(x_1, x_2) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$
   $$-6 \le x_1 \le 6, \quad -6 \le x_2 \le 6$$
      i. Based on your experience in above experiment use the best combination of the crossover and mutation probability for real coded GA using:
         A. Simulated Binary Crossover (SBX)
         B. Polynomial Mutation (PM)
      ii. Compare with Binary GA.
   (b) Use:

i. Binary GA
ii. Real-Coded GA

(c) Perform 30 runs and report:

Method	Best	Worst	Average	Std. Dev.
Binary GA				
Real GA				

5. **Constrained Genetic Algorithm** Consider the problem Minimize:

$$f(\mathbf{x}) = (x_1^2 + x_2 - 11)^2 + (x_1 + x_2^2 - 7)^2$$

Subject to:

$$g_1(\mathbf{x}) = 4.84 - (x_1 - 0.05)^2 - (x_2 - 2.5)^2 \geq 0$$
$$g_2(\mathbf{x}) = x_1^2 + (x_2 - 2.5)^2 - 4.84 \geq 0$$

Bounds:

$$0 \leq x_1 \leq 6, \qquad 0 \leq x_2 \leq 6$$

Use the parameter free penalty method for both real and binary coded GA. Perform 30 runs and report:

Method	Best	Worst	Average	Std. Dev.
Binary GA				
Real GA				

6. **Submission**

(a) Report (plots, tables, and short discussion on your observation.)